

NATURAL LANGUAGE PROCESSING FOR LEGAL TEXT ANALYSIS



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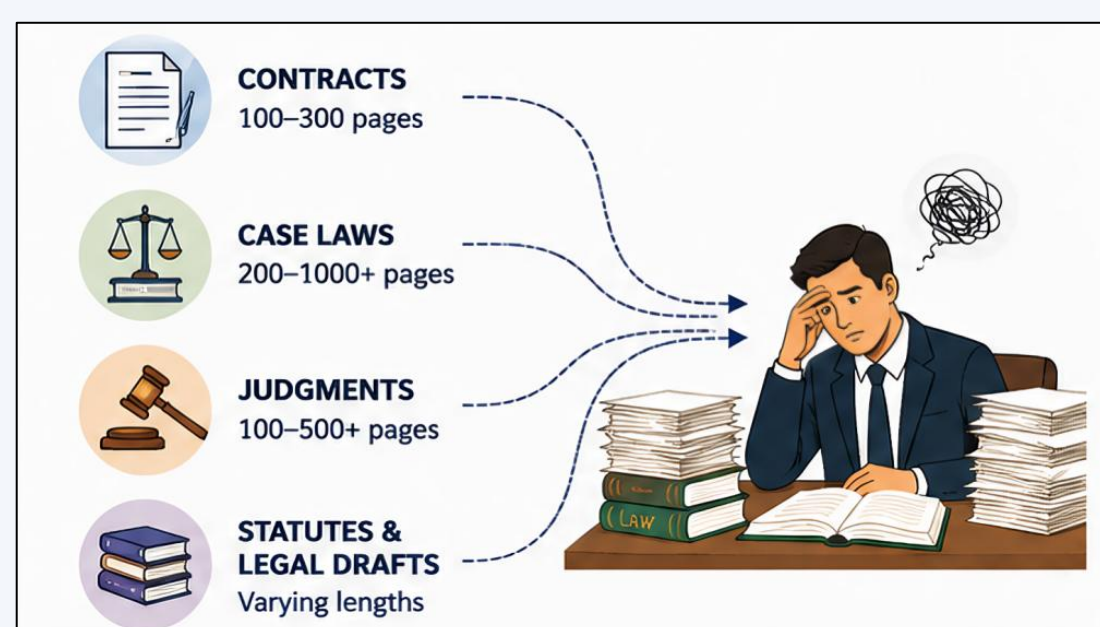
MENTORS

Dr. Saptarshi Ghosh (IIT KGP) and Dr. Kripabandhu Ghosh (IISER Kolkata)

1. MOTIVATION & PROBLEM

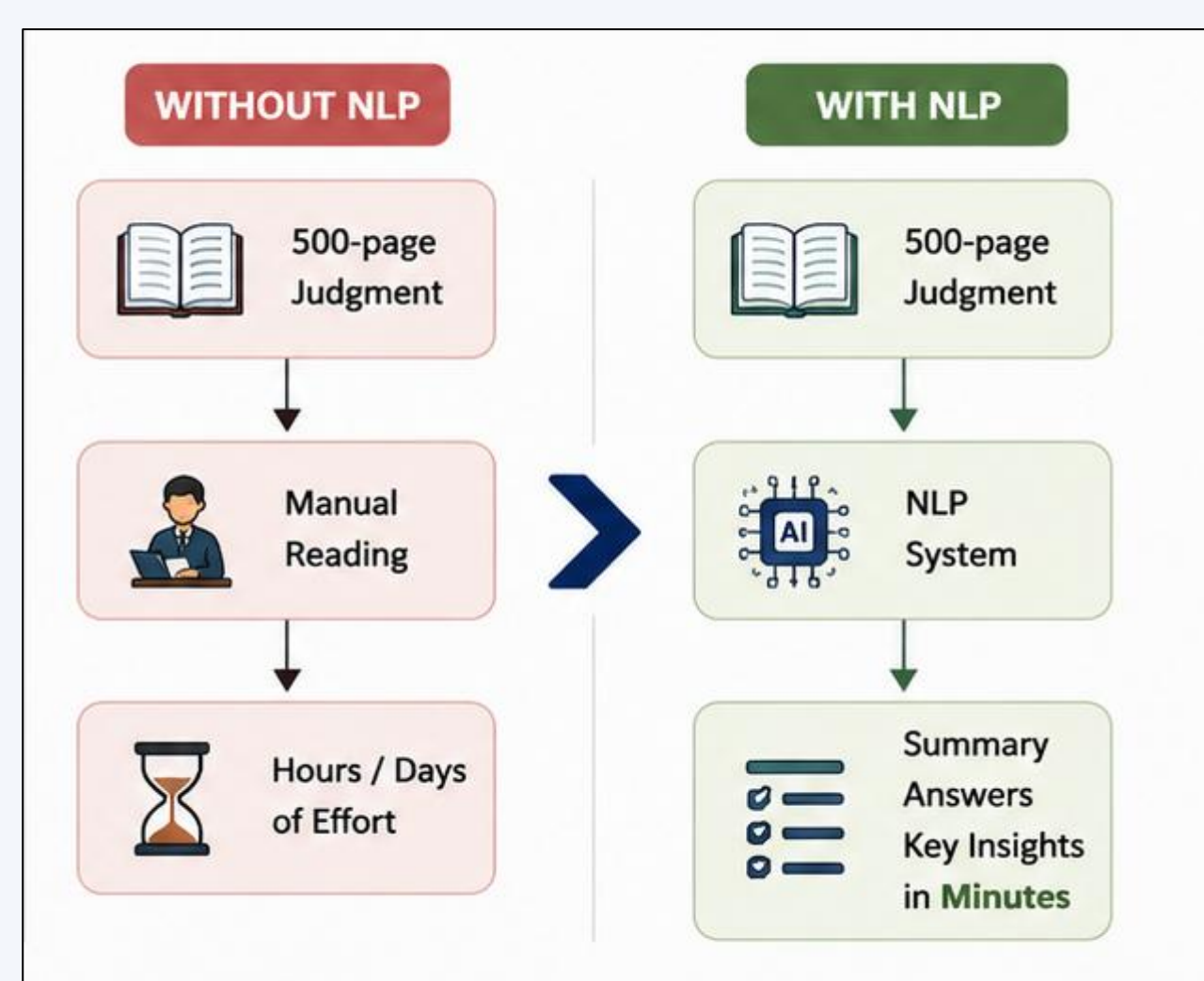
1. Problem

- Legal documents are lengthy and complex (often hundreds of pages).
- Manual review is time-consuming.
- Important information may be difficult to locate.
- Impossible to understand at one read.



2. Why NLP?

- Reads the entire document and generates a comprehensive summary.
- Can be used for tasks like Q&A, information retrieval and summarization.

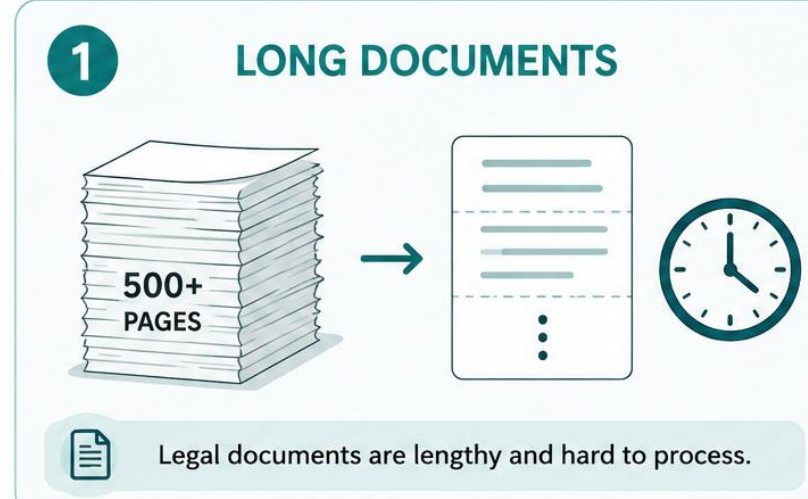


3. CHALLENGES

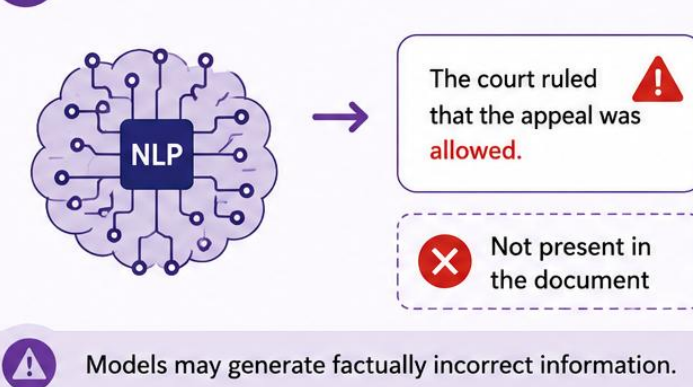
Although NLP has significant potential and capabilities, there are some problems in applying it to the legal domain.

1. LARGE DOCUMENTS

Processing large documents takes a significant amount of time of computational resources.



2. HALLUCINATIONS

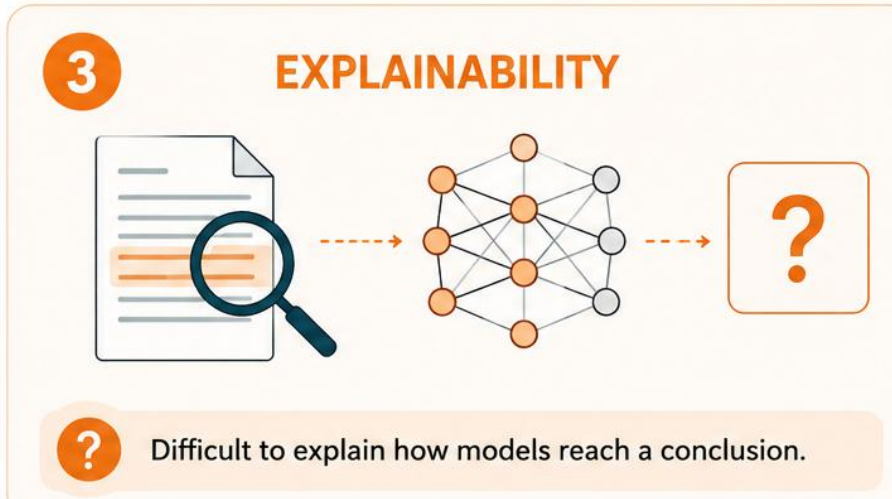


2. HALLUCINATIONS

If there is some ambiguity in the text, the model can generate false information based on what it has learnt.

3. EXPLAINABILITY

It is difficult to interpret the predictions of the model – how it reached a certain conclusion.



4. PRIVACY & SECURITY



4. PRIVACY CONCERNS

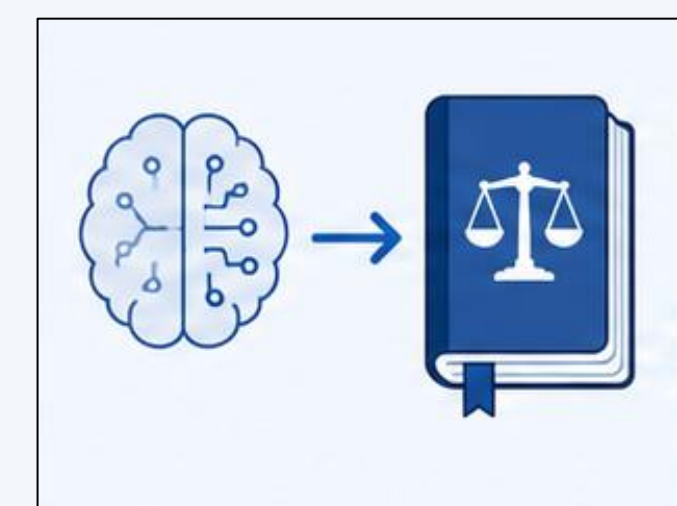
Since legal data contains sensitive information, it must be protected from being misused by the model.

5. FUTURE DIRECTIONS

Currently, there is work ongoing in the field of applying NLP to the legal domain. The current works are in the following topics:

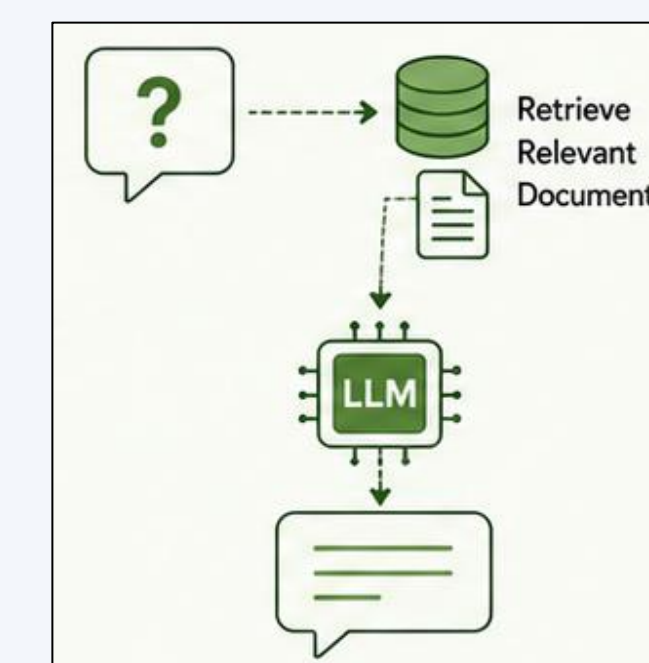
1. BETTER LEGAL LLMS

- Models trained specifically on legal corpora.
- Better performance on legal tasks.



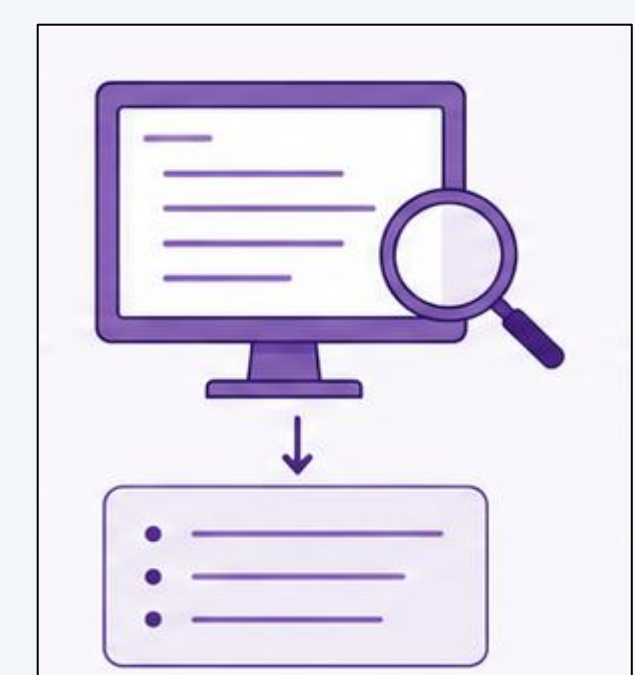
2. RETRIEVAL AUGMENTED GENERATION (RAG)

- Retrieve relevant legal document.
- Generate answers using verified sources.
- Reduce hallucinations.



3. EXPLAINABLE AI

- Understand why a model produced an answer.
- Improve transparency and trust.



4. MULTILINGUAL AI

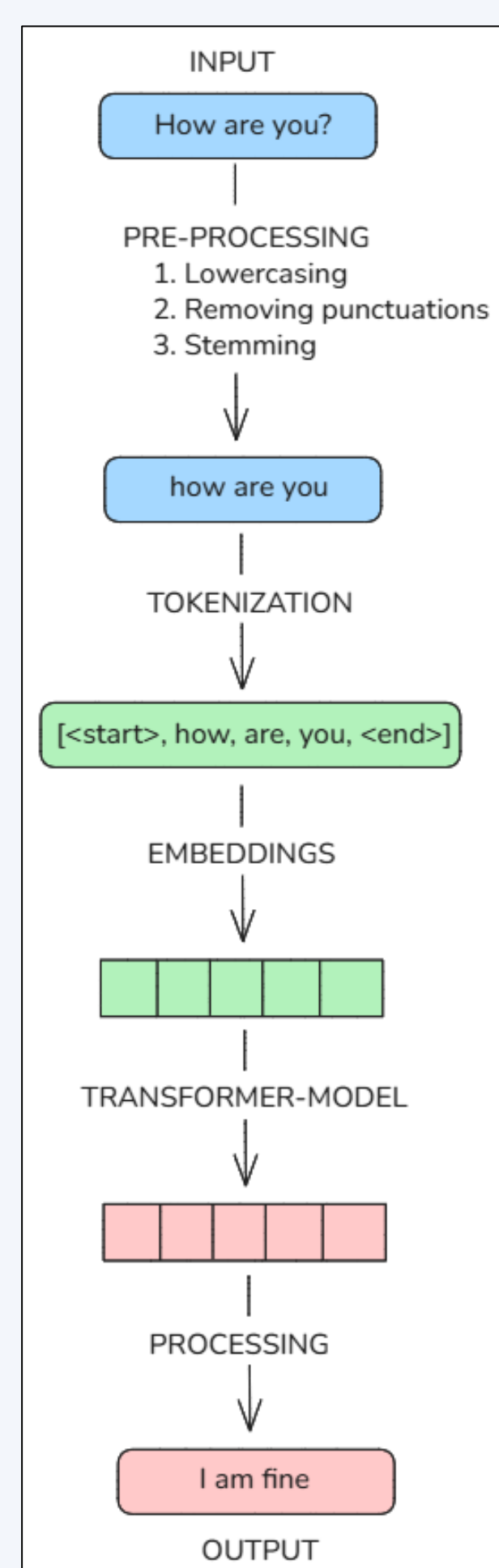
- Support Indian language.
- Improve accessibility.
- Enable wider adoption.



2. PROCESSING & APPLICATIONS

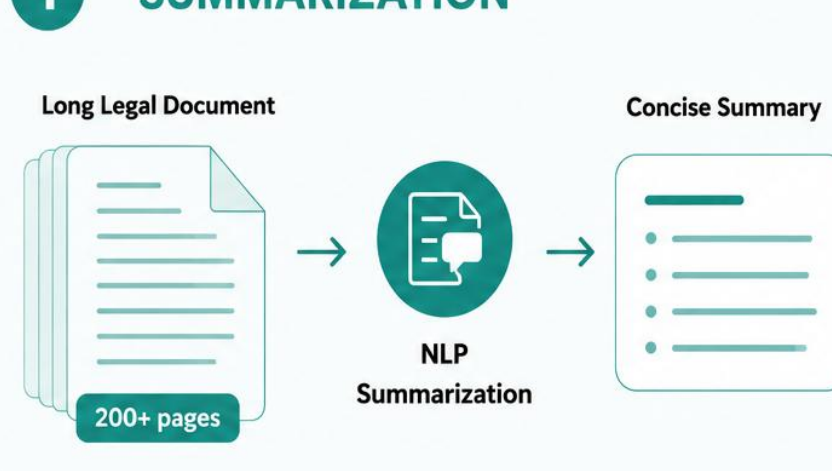
1. NLP Processing

The legal documents are given as input to a Transformer-based NLP model. It processes them as follows:

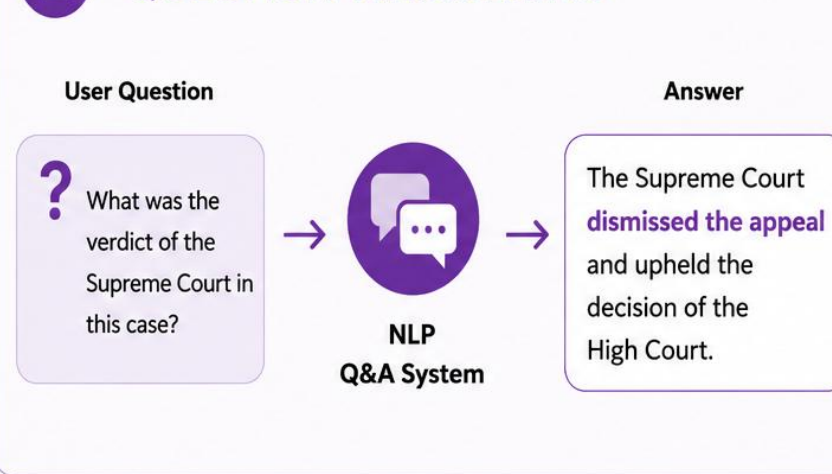


2. Applications

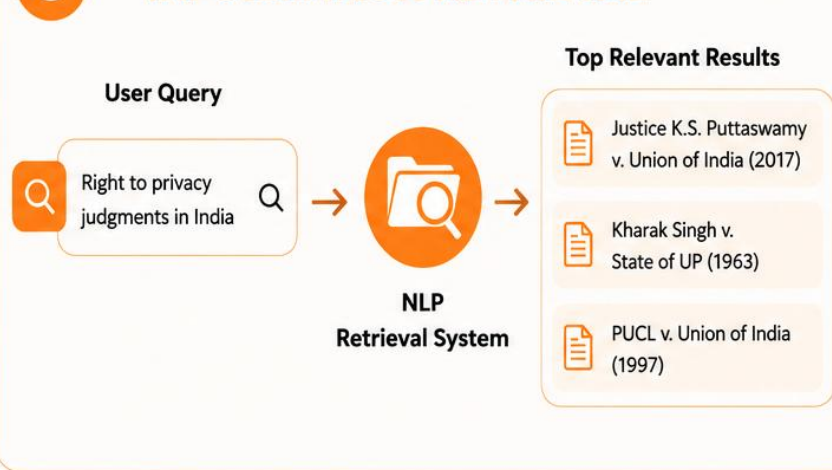
1. SUMMARIZATION



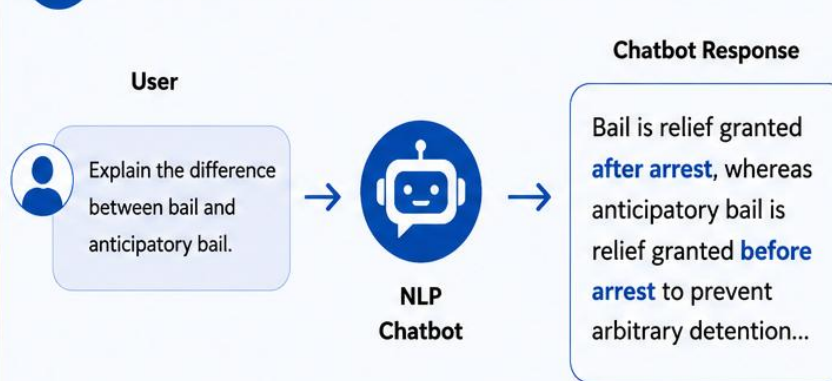
2. QUESTION-ANSWERING



3. INFORMATION RETRIEVAL



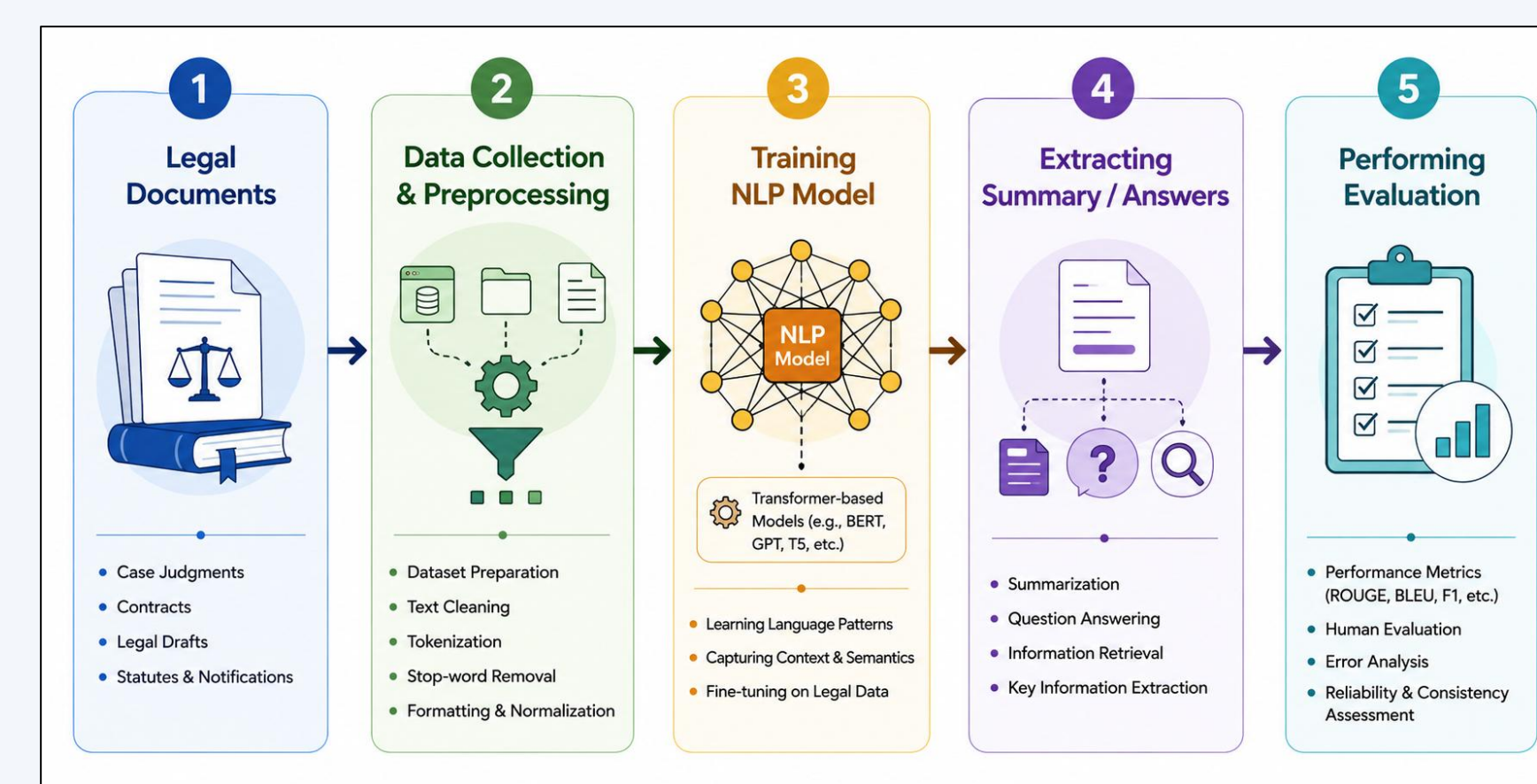
4. LEGAL ASSISTANCE CHATBOT



4. MY INTERNSHIP WORK

GRISHMA Summer Internship, IIT Kharagpur

1. Project Workflow



2. Research Objectives

- Summarizing legal documents.
- Developing a question answering system on the legal text.
- Mitigating hallucinations.
- Developing an information retrieval system.
- Improving the interpretability of the model.

3. Current Progress

- Studied and implemented traditional sequential models like RNN, LSTM, Encode-decoder sequences, etc on public datasets.
- Studied transformers and their architecture.
- Studied how to interpret simple models.
- Currently exploring NLP techniques for legal summarization, question answering, and hallucination mitigation.

6. CONCLUSION

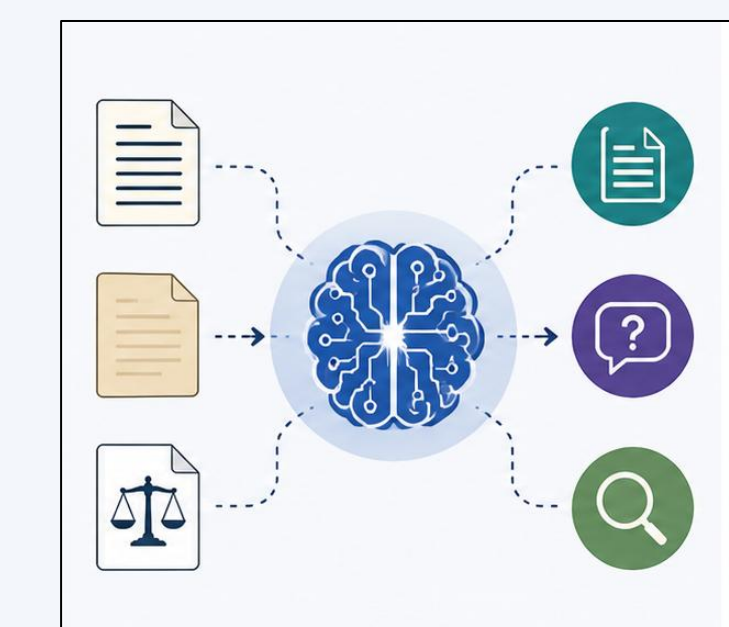
Problem

Legal NLP addresses the challenge of analyzing large and complex legal documents.



Opportunity

Transformer-based language models enable summarization, question answering, and information retrieval.



Future Challenge

Building reliable, explainable, and trustworthy legal AI remains an important research direction.



REFERENCES:

- [1] Vaswani, A. et al. *Attention Is All You Need*
- [2] Alammur, J. *The Illustrated Transformer*
- [3] CampusX YouTube channel *Transformers playlist*

Scan to view currently ongoing work (GitHub repository)

